A smaller counterexample to the mixed-integer $1/e$ bound

52 integer variables and a Boolean family of outer fibers

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Abstract

We give a rational convex polytope whose mixed-integer halfspace depth is strictly less than $1/(e2^{52})$ at every mixed-integer point. Its continuous dimension is $274784 + 2^{61}$. The improvement comes from adding entire Boolean families of outer fibers, while preserving the origin fiber by explicit affine inequalities. This is about 2^{32} times smaller in continuous dimension than the earlier $n = 76$, $d = 1372131 + 2^{93}$ example, but it remains far above a billion variables. The argument below and its exact arithmetic certificate have been independently reviewed; this new construction has not been formalized in Lean.

1 Statement and the two inequalities that drive the proof

For a bounded set $S \subset \mathbb{Z}^n \times \mathbb{R}^d$ with measurable fibers, write

$$\mu(S) = \sum_{z \in \mathbb{Z}^n} \mathrm{vol}_d(S_z), \qquad \mathrm{depth}_S(y) = \inf_{\substack{H \text{ closed ambient halfspace} \\ y \in H}} \frac{\mu(S \cap H)}{\mu(S)}.$$

The depth is defined when $0 < \mu(S) < \infty$. Halfspaces are proper: their normal is nonzero. In our example every nonempty fiber is a full-dimensional convex body, so μ agrees with the standard Euclidean Hausdorff measure $\mathcal{H}_d$ on S.

Theorem 1. *There is a rational convex polytope C with nonempty interior in $\mathbb{R}^{52} \times \mathbb{R}^d$, where*

$$d = 274784 + 2^{61} = 2305843009213968736,$$

such that $S = C \cap (\mathbb{Z}^{52} \times \mathbb{R}^d)$ is nonempty and

$$\sup_{y \in S} \mathrm{depth}_S(y) < \frac{1}{e2^{52}}.$$

Every nonempty integer fiber has dimension d. In particular this also violates $2^{-52}(d/(d+1))^d$.

This addresses the specific constant $1/e$. It does not rule out a smaller positive absolute constant in the weak conjecture.

Here is the mass bookkeeping before the geometry. We first construct a body $C_0 \subset \mathbb{R}^{52} \times \mathbb{R}^{3+h}$. Normalize volumes by the volume b of its origin fiber. The $2^{52} - 1$ positive Boolean fibers all have mass m. The outer fibers have total mass E, split equally among four groups. Thus its total mass is bW, where

$$W = (2^{52} - 1)m + 1 + E = 2^{52}m + \Delta, \qquad \Delta = 1 + E - m. \tag{1}$$

Four halfspaces cover the origin fiber; each has normalized mass $L = 27/64 + E/4$ and misses all positive Boolean fibers. We arrange

$$L < \frac{m}{e}, \qquad \Delta > 0. \tag{2}$$

These cuts handle every point in the origin fiber. Every other integer base is exposed, and its fiber has mass at most m. A common simplex factor of dimension ℓ supplies a small cap at every point of those fibers. It suffices to choose

$$\frac{2^{52} m}{2\ell - 1} < \Delta. \tag{3}$$

Section 5 explains this last step, including isolation by a closed ambient halfspace. No rounding of a center is used.

2 The fibers and their integer bases

Let $B_0 \subset \mathbb{R}^3$ be the standard tetrahedron translated so that its centroid is zero. Its barycentric coordinates $\lambda_0, \dots, \lambda_3$ satisfy $\lambda_j(0) = 1/4$. Put $f_j = \lambda_j - 1/4$; these are linear and $\sum_j f_j = 0$. Define

$$T_a = \{x \in B_0 : \lambda_j(x) \le 1/4 \ (j \ne a)\}, \qquad \theta = \frac{\mathrm{vol}(T_a)}{\mathrm{vol}(B_0)} = \frac{3}{32}, \qquad \beta = \frac{27}{64}.$$

Indeed, the three coordinates λ_j, $j \ne a$, identify T_a with $[0, 1/4]^3$ and B_0 with a simplex of volume $1/6$; the Jacobian cancels in the ratio. Also $\{f_j \ge 0\} \cap B_0$ is a tetrahedron of linear scale $3/4$, hence volume fraction β.

Partition the 52 integer coordinates into four blocks $I_0, \dots, I_3$ of size $N = 13$. For nonempty $I \subset \{1, \dots, 52\}$, let $a(I)$ be the block of its least member. The intended integer support is

$$Z = \{0,1\}^{52} \ \cup \ \{-\mathbf{1}_J : \varnothing \ne J \subset I_a, \quad a = 0, 1, 2, 3\}. \tag{4}$$

The added bases include all nonempty subsets of a block, rather than only the negative unit vectors. There are $4(2^{13} - 1)$ of them.

Choose the following rational parameters:

$N = 13$	$h = 274781$	$\ell = 2^{61}$
$s = 500000001/10^9$	$t = 999999997/10^9$	$q = 100001/200000$
$\eta = 399999999/400000000$	$\varepsilon = 1/10^{12}$	

Let c_a have barycentric coordinates $1/4 + 3\varepsilon$ at a and $1/4 - \varepsilon$ elsewhere. Write

$$F = [-1, 1]^h, \quad B = B_0 \times F, \quad \delta_s = 2s - 1, \quad \delta_t = 1 - t, \quad \delta_q = 2q - 1.$$

For $k \ge 1$ and $1 \le r \le 13$, set

$$\begin{aligned} K(a,k) &= \big[2(c_a + s(T_a - c_a)) + (k-1)c_a\big] \times (2qF), \\ G(a,r) &= \big[t_r(T_a - c_a) - (r-1)c_a\big] \times (u_r F), \\ t_r &= 1 - r\delta_t, \qquad u_r = 1 - r\delta_q. \end{aligned}$$

Define C_0 to be the convex hull of the following assigned fibers:

Integer base	Assigned fiber	Volume divided by $b = \mathrm{vol}(B)$
0	B	1
$\mathbf{1}_I$, $I \neq \varnothing$	$K(a(I), \lvert I\rvert)$	$m = \theta(2s)^3(2q)^h$
$-\mathbf{1}_J$, $\varnothing \neq J \subset I_a$	$G(a, \lvert J\rvert)$	$\alpha_{\lvert J\rvert} = \theta t^3_{\lvert J\rvert} u^h_{\lvert J\rvert}$

The proof must check that convexification neither creates extra integer bases nor enlarges the assigned origin fiber. We do that explicitly next.

3 Why these are exactly the actual fibers

Lemma 1. *The integer support of C_0 is exactly (4). All its nonzero integer bases are exposed and their actual fibers are the assigned ones. Its origin fiber is exactly B. The body C_0 is a rational polytope with nonempty ambient interior, and all its nonempty fibers have dimension $3 + h$.*

Proof. Every projected coordinate lies in $[-1, 1]$. If an integer projection has coordinate -1 at an index in I_a, every generator in a convex representation has that coordinate -1. Such generators all belong to the negative Boolean block I_a. Thus the projection is a negative Boolean vector in that block. If no coordinate is negative, it is Boolean. Every base listed in (4) was included, proving the census.

For exposure, put $\rho = 1/53$. The positive base $\mathbf{1}_I$ uniquely maximizes the functional with coefficients 1 on I and $-\rho$ elsewhere: its value is $|I| \geq 1$, whereas every negative base has value at most $13\rho < 1$. Similarly $-\mathbf{1}_J$ uniquely maximizes the functional with coefficients -1 on J and ρ elsewhere. Other negative bases have smaller value, and positive bases have value at most $\rho(52 - |J|) < 1 \leq |J|$. An exposed base can only use generators above it in a convex representation. Its fiber is therefore exactly the assigned convex fiber.

For the origin, define

$$v = \delta_s/4 + (1 - \delta_s)\varepsilon.$$

Every generator (z, x, w) satisfies the affine inequalities

$$\lambda_j(x) + v \sum_i z_i \geq 0 \quad (j = 0, \ldots, 3), \qquad \pm w_l \leq 1 + \delta_q \sum_i z_i \quad (l = 1, \ldots, h). \tag{5}$$

Here are the details. At a positive base of size k and selected type a,

$$\min \lambda_j = \begin{cases} -\delta_s/4 - (k - \delta_s)\varepsilon, & j \neq a, \\ 1/4 + 3(k - \delta_s)\varepsilon, & j = a. \end{cases}$$

The first value is at least $-kv$, since the difference is $(k - 1)\delta_s(1/4 - \varepsilon) \geq 0$, and the second is positive. The box bound follows from $2q = 1 + \delta_q \leq 1 + k\delta_q$.

At a negative base of size r in block a, the corresponding minima are

$$\min \lambda_j = \begin{cases} r(\delta_t/4 + t\varepsilon), & j \neq a, \\ 1/4 - 3rt\varepsilon, & j = a. \end{cases}$$

The first exceeds rv by $r(\delta_t - \delta_s)(1/4 - \varepsilon) > 0$. The second is at least rv because our parameters satisfy

$$N[\delta_s + 4\varepsilon(3t + 1 - \delta_s)] \leq 1. \tag{6}$$

The free-box bound is precisely $u_r = 1 - r\delta_q$. The origin generators also satisfy (5).

These inequalities persist under convex combinations. At $z = 0$, they say $x \in B_0$ and $w \in F$, so the origin fiber is contained in B. The reverse inclusion holds by construction. This argument includes all convex combinations of overlapping negative subsets.

All parameters and vertices are rational, and $t_r, u_r > 0$, so all assigned fibers are full-dimensional rational polytopes. The origin fiber spans the vertical directions, while the positive unit bases span all integer directions. Consequently C_0 has full affine span and nonempty interior. □

4 The four cuts at the origin

Put $R = 159 = 3(52 + 1)$. For $j = 0, \dots, 3$, define the closed halfspace

$$H_j = \left\{ (z, x, w) : f_j(x) - R \sum_{i \in I_j} z_i + \eta\varepsilon \sum_{i \notin I_j} z_i \geq 0 \right\}.$$

It ignores the free-box coordinates. These four halfspaces cover the origin fiber because $\sum_j f_j = 0$.

On a positive fiber of size k that meets I_j, the continuous score is at most $k+1$ and the integer score at most $-R+k-1$. Their sum is negative. If its base avoids I_j, its selected type differs from j and the maximum total score is

$$[k\eta - (k - \delta_s)]\varepsilon < 0,$$

since $\eta < 1 - \delta_s$. Thus H_j misses every positive Boolean fiber.

On a negative fiber $G(a, r)$ with $a \neq j$, the maximum score is $r(t - \eta)\varepsilon < 0$. On $G(j, r)$ the minimum score is $rR - 3rt\varepsilon > 0$, so the whole fiber is included. Exactly one negative block is captured. It follows that, with

$$E = 4 \sum_{r=1}^{13} \binom{13}{r} \alpha_r,$$

the total volume is bW from (1), and

$$\mu\big((C_0 \cap (\mathbb{Z}^{52} \times \mathbb{R}^{3+h})) \cap H_j\big) = b \left(\frac{27}{64} + \frac{E}{4} \right) = bL. \tag{7}$$

The normals are nonzero. All omitted fibers have strict score gaps, and the origin cap boundaries have zero volume, so this equality holds for the closed halfspaces as stated.

5 A small cap at every other point

Lemma 2. *For every positive integer ℓ, put $\Gamma_\ell = (\ell/(\ell+1))^\ell$. Then*

$$e\Gamma_\ell < 1 + \frac{1}{2\ell - 1}.$$

Every point of an ℓ-simplex lies in a closed cap of relative volume at most Γ_ℓ.

Proof. The exact identity

$$\log(e\Gamma_\ell) = \int_0^1 \frac{t}{\ell + t}\, dt < \frac{1}{2\ell}$$

and $e^x < 1/(1-x)$ for $0 < x < 1$ give the first assertion. The latter inequality follows by integrating $1/(1-t) > 1$ from 0 to x. For the second, choose a largest barycentric coordinate a of the point.

Then $a \ge 1/(\ell+1)$. The cap where that coordinate is at least a has relative volume $(1-a)^\ell \le \Gamma_\ell$, including zero volume when $a = 1$. □

Let J be the standard rational ℓ-simplex and set $C = C_0 \times J$. Every nonzero fiber of C_0 has volume at most mb, because $\alpha_r \le \theta < m$. For a point of C above a nonzero integer base z, take the cap $A(v') \ge c$ supplied by Lemma 2 in the J coordinate. Let g expose z among the finitely many integer bases. For sufficiently large finite $Q > 0$, the halfspace

$$A(v') - c + Q\big(g(z') - g(z)\big) \ge 0$$

agrees with the cap above z and misses all other fibers. Indeed, all exposure gaps are strictly negative and $A - c$ is bounded on compact J. Its normal is nonzero because the simplex-cap functional is nonzero. Its mass is at most $mb\Gamma_\ell \operatorname{vol}(J)$.

For a point above zero, one of the four cuts from the previous section, extended to ignore J, has mass $bL \operatorname{vol}(J)$. Thus every point in the actual mixed-integer set has a containing closed halfspace of relative mass at most

$$\frac{\max\{L, \Gamma_\ell m\}}{W}. \tag{8}$$

By (2), $L/W < m/(eW) < 1/(e2^{52})$. By (3) and Lemma 2,

$$e\Gamma_\ell\, 2^{52} m < 2^{52} m + \frac{2^{52} m}{2\ell - 1} < 2^{52} m + \Delta = W.$$

This bounds the second term in (8) strictly below the same target. Only the exact parameter checks remain.

6 Exact inequalities for the displayed parameters

All geometric hypotheses used above follow directly from the rational parameters. The nontrivial positive margins are

$$\eta - t = 2(1-s) - \eta = \frac{1}{2 \cdot 10^9},$$
$$1 - 13\delta_t = \frac{999999961}{10^9}, \qquad 1 - 13\delta_q = \frac{99987}{100000},$$
$$1 - 13[\delta_s + 4\varepsilon(3t + 1 - \delta_s)] = \frac{249999993448000000143}{250000000000000000000}.$$

In particular $0 < \varepsilon < 1/4$, $1/2 < s, q < 1$, $0 < t < \eta < 2(1-s)$, $t_r, u_r > 0$, and (6) hold.

The exact rational certificate described below proves

$$\frac{1463268}{10^6} < m < \frac{1463269}{10^6}, \qquad \frac{465669}{10^6} < E < \frac{465670}{10^6}. \tag{9}$$

Put $U = 1359141/500000$. The exponential series gives

$$e < \sum_{j=0}^{8} \frac{1}{j!} + \frac{10}{9 \cdot 9!} < U,$$

because the tail after $1/9!$ has successive ratios at most $1/10$, with later ratios strictly smaller. The margin in the second inequality is $1717/10206000000$. Using (9),

$$\frac{1463268/10^6}{U} - \frac{27}{64} - \frac{465670/10^6}{4} = \frac{40641}{2970800000} > 0,$$

$$\Delta > 1 + \frac{465669}{10^6} - \frac{1463269}{10^6} = \frac{3}{1250} > 0,$$

$$\frac{3}{1250} - \frac{2^{52}(1463269/10^6)}{2\ell - 1} = \frac{139939647533841645493}{14411518807585587196875 0} > 0.$$

These are precisely (2) and (3). They complete the proof of Theorem 1. The dimension is $d = 3 + h + \ell = 274784 + 2^{61}$. Compactness, full-dimensionality, and the fiber census were proved before adjoining J and persist under this product. All fibers have finite positive d-volume, so the intrinsic Hausdorff dimension is d and its measure is their finite volume sum. Finally $\log(1 + 1/d) < 1/d$ gives $(d/(d+1))^d > 1/e$.

Reproducing the power certificate

The companion files, in the same repository directory as this source, are

```
check_integer_dimension_52.py
integer-dimension-52-certificate.json
```

The program uses integer arithmetic and `fractions.Fraction`; no floating-point estimates are premises. It encloses the fourteen powers $(2q)^h$ and u_r^h, $1 \le r \le 13$, at fixed denominator $D = 10^{40}$.

For a nonnegative rational number x, initialize bounds $\lfloor Dx \rfloor / D \le x \le \lceil Dx \rceil / D$. If integer endpoint pairs $[L, U]$ and $[L', U']$ enclose two nonnegative quantities after division by D, their product is enclosed by

$$\left[\frac{\lfloor LL'/D \rfloor}{D}, \frac{\lceil UU'/D \rceil}{D} \right].$$

Monotonicity of multiplication proves this assertion. Apply it at every multiplication and squaring in binary exponentiation. For the underlying exact values, the invariant is accumulator $\cdot\, \text{base}^r = x^h$, where r is the remaining exponent; the stored intervals enclose these values. For an odd r, multiply the accumulator by the base, then halve the exponent after subtracting one; for an even r, halve it. Square the base whenever a further iteration is needed. Induction proves that the final accumulator interval encloses the exact power.

The JSON records the exponent and all interval endpoints at every step. Multiplication by the positive exact factors $\theta(2s)^3$ and θt_r^3, followed by the positive binomial sum, yields rational bounds strictly inside (9). The program also verifies every displayed scalar margin. This is a finite exact arithmetic certificate; it is separate from the geometric proof above and from Lean verification of the earlier $n = 76$ construction.

AI disclosure

The construction and proof in this note were obtained mainly by AI research agents, using a fact graph system similar to Danus (arXiv:2607.06447). The result is fact `df488edae05a1702` of the weak Oertel conjecture fact graph,[1] where it carries the label `LLM-verified`: language models reviewed the proof, and the label does not assert human review. The result is not formalized in Lean. This note was written with AI assistance.

[1] https://github.com/Hongyu-Cheng/weak-oertel-conjecture